Performance Policies and Alerts

Cloud Insights

NetApp
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Performance Policies and Alerts

Alerting with Monitors

You create monitors to set thresholds that trigger alerts to notify you about issues related to the resources in your network. For example, you can create a monitor to alert for *node write latency* for any of a multitude of protocols.

Monitors and Alerting is available in Cloud Insights Standard Edition and higher.

When the monitored threshold and conditions are reached or exceeded, Cloud Insights creates an alert. A Monitor can have a *Warning* threshold, a *Critical* threshold, or both.

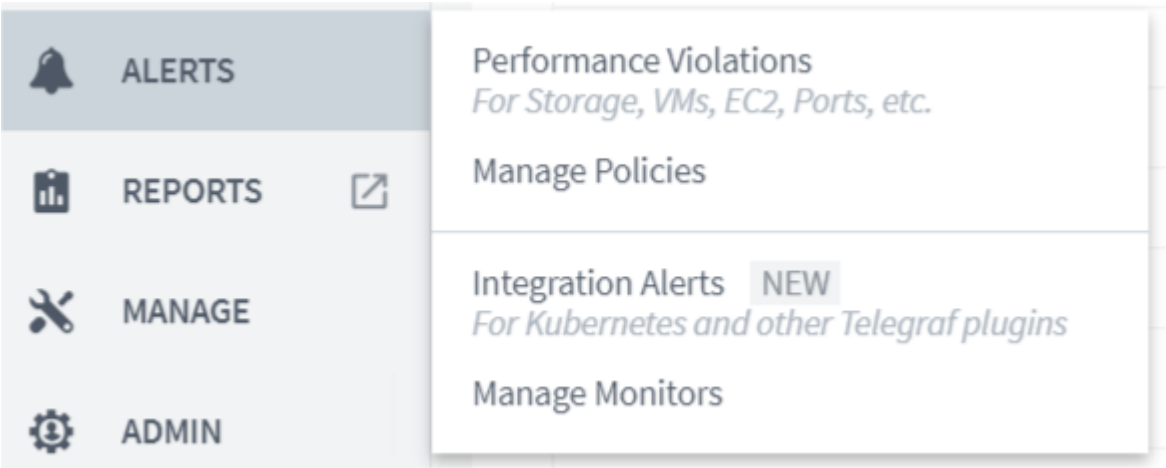
Monitor or Performance Policy?

What's the difference between a **Performance Policy** and a **Monitor**?

Policies allow you to set thresholds on "infrastructure" objects such as storage, VM, EC2, and ports. These policies trigger violations when thresholds are met or exceeded. Each violation can be investigated for troubleshooting. Policies are described in detail elsewhere in this documentation.

Monitors provide similar functionality for "integration" data such as those collected for Kubernetes, ONTAP advanced metrics, and Telegraf plugins, and alert when thresholds are crossed. With Monitors, you can set thresholds for Warning- or Critical-level alerts, or both.

Policies and Monitors are available under the **Alerts** menu.



Emails can be sent when a policy or monitor is triggered.

Creating a Monitor

In the example below, we will create a Monitor to give a Warning alert when *Volume Node NFS Write Latency* reaches or exceeds 200ms, and a Critical alert when it reaches or exceeds 400ms. We only want to be alerted when either threshold is exceeded for at least 15 continuous minutes.

Requirements

- Cloud Insights must be configured to collect integration data, and that data is being collected.

Create the Monitor

1. From the Cloud Insights menu, click **Alerts > Manage Monitors**

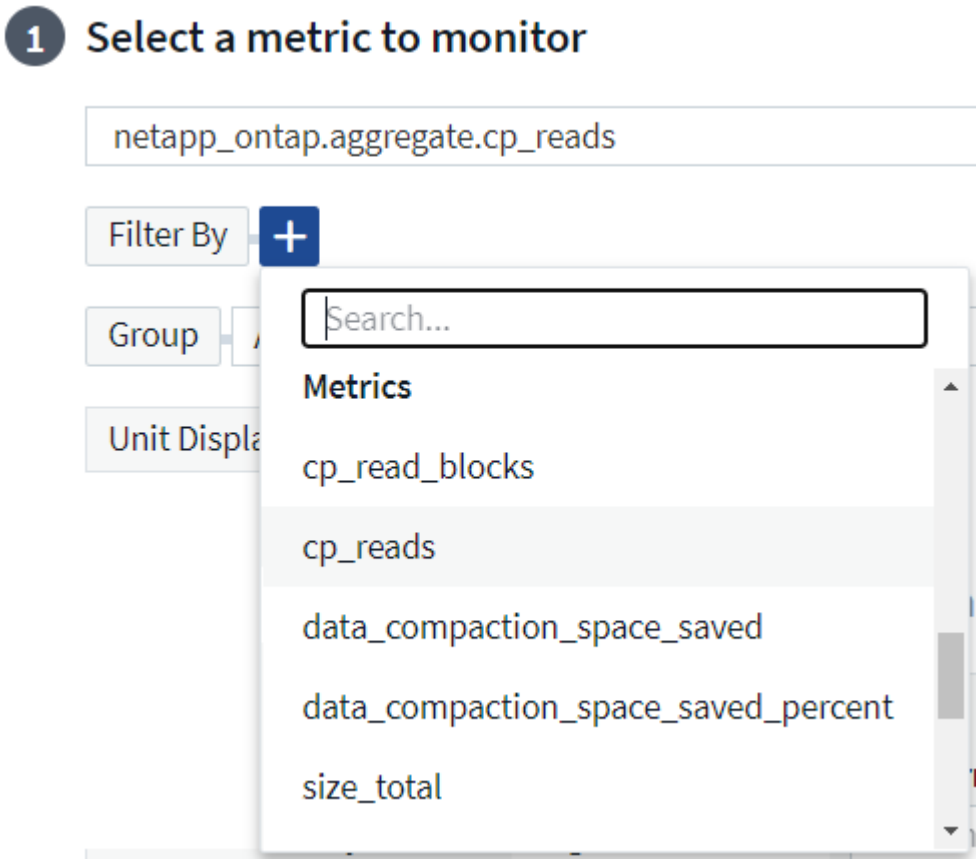
   The Monitors list page is displayed, showing currently configured monitors.

2. To add a monitor, Click **+ Monitor**. To modify an existing monitor, click the monitor name in the list.

   The Monitor Configuration dialog is displayed.

3. In the drop-down, search for and choose an object type and metric to monitor, for example *netapp_ontap_volume_node_nfs_write_latency*.

You can set filters to narrow down which object attributes or metrics to monitor.



When working with integration data (Kubernetes, ONTAP Advanced Data, etc.), metric filtering removes the individual/unmatched data points from the plotted data series, unlike infrastructure data (storage, VM, ports etc.) where filters work on the aggregated value of the data series and potentially remove the entire object from the chart.

To create a multi-condition monitor (e.g., IOPS > X and latency > Y), define the first condition as a threshold and the second condition as a filter.

Define the Conditions of the Monitor.

1. After choosing the object and metric to monitor, set the Warning-level and/or Critical-level thresholds.
2. For the *Warning* level, enter 200. The dashed line indicating this Warning level displays in the example graph.
3. For the *Critical* level, enter 400. The dashed line indicating this Critical level displays in the example graph.

   The graph displays historical data. The Warning and Critical level lines on the graph are a visual representation of the Monitor, so you can easily see when the Monitor might trigger an alert in each case.

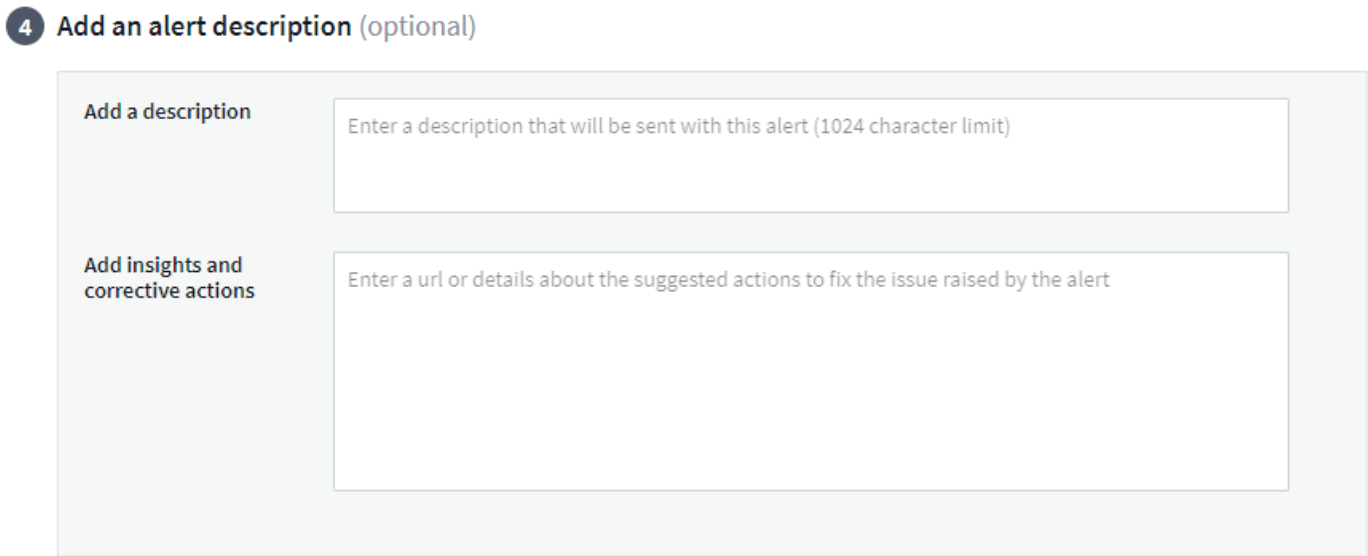
4. For the occurence interval, choose *Continuously* for a period of *15 Minutes*.

   You can choose to trigger an alert the moment a threshold is breached, or wait until the threshold has been in continuous breach for a period of time. In our example, we do not want to be alerted every time the Total IOPS peaks above the Warning or Critical level, but only when a monitored object continuously exceeds one of these levels for at least 15 minutes.

Setting Corrective Actions or Additional Information

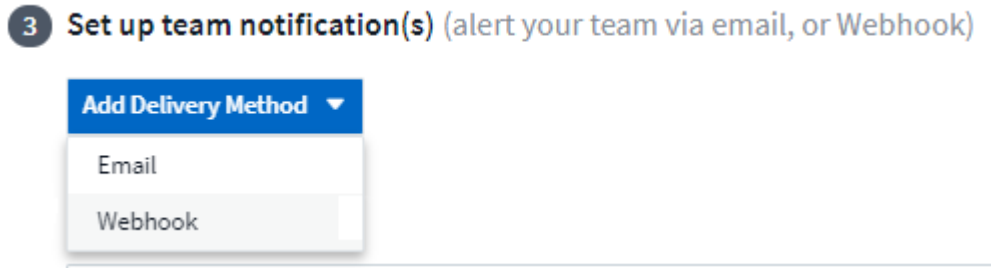
You can add an optional description as well as additional insights and/or corrective actions by filling in the **Add an Alert Description** section. The description can be up to 1024 characters and will be sent with the alert. The insights/corrective action field can be up to 67,000 characters and will be displayed in the summary section of the alert landing page.

In these fields you can provide notes, links, or steps to take to correct or otherwise address the alert.



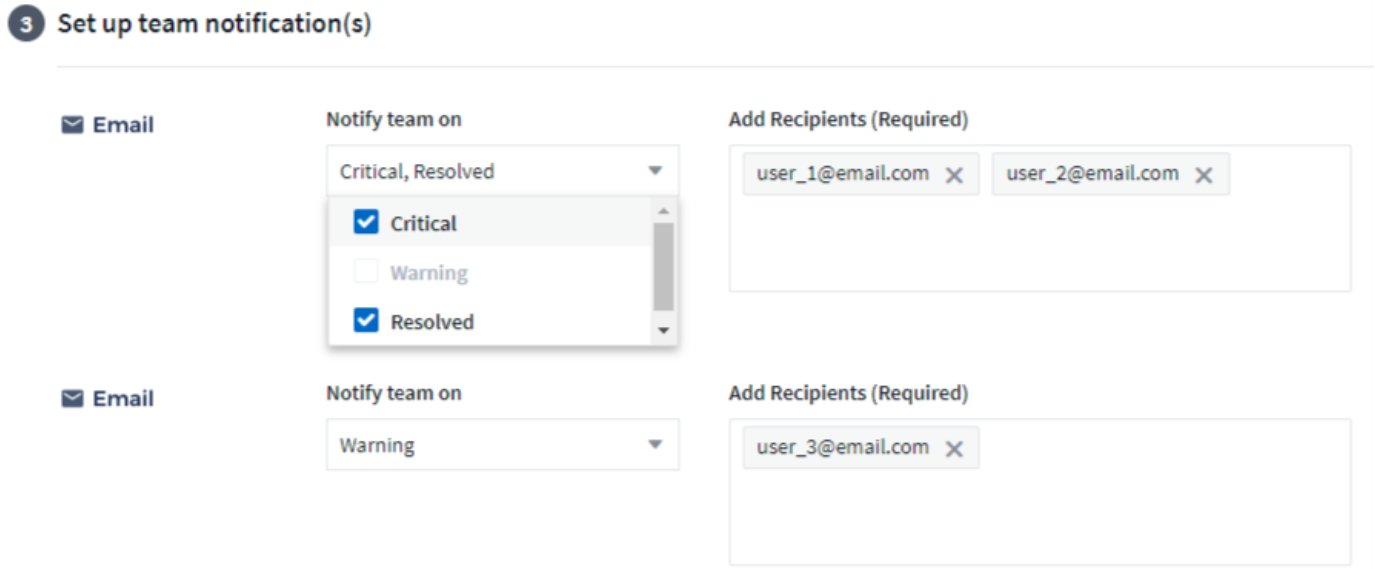
Select notification type and recipients

In the *Set up team notification(s)* section, you can choose whether to alert your team via email or Webhook.



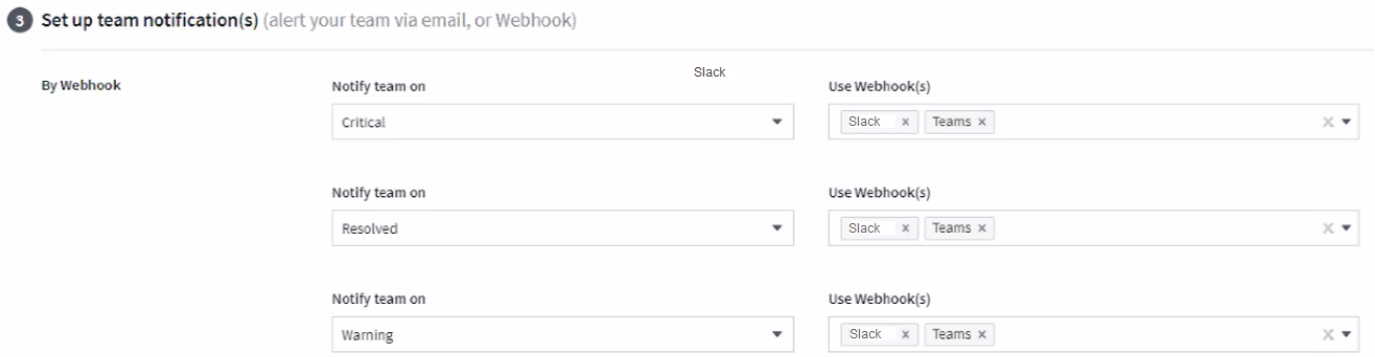
Alerting via Email:

Specify the email recipients for alert notifications. If desired, you can choose different recipients for warning or critical alerts.



Alerting via Webhook:

Specify the webhook(s) for alert notifications. If desired, you can choose different webhooks for warning or critical alerts.



Webhooks is considered a Preview feature and is therefore subject to change.

Save your Monitor

1. If desired, you can add a description of the monitor.
2. Give the Monitor a meaningful name and click **Save**.

   Your new monitor is added to the list of active Monitors.

Monitor List

The Monitor page lists the currently configured monitors, showing the following:

- Monitor Name
- Status
- Object/metric being monitored
- Conditions of the Monitor

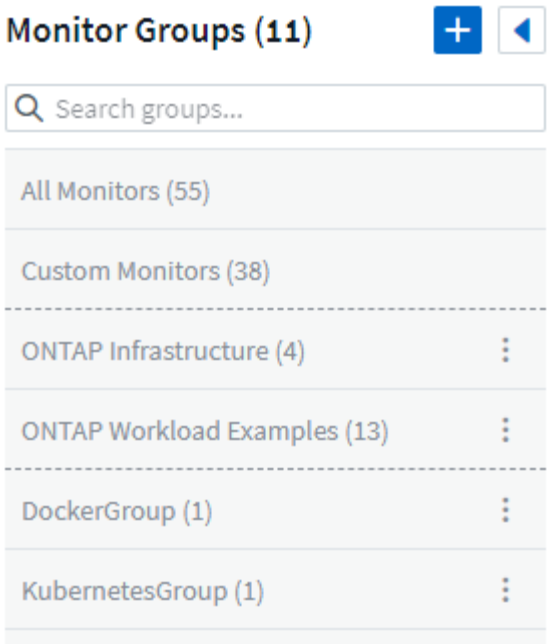
You can choose to temporarily suspend monitoring of an object type by clicking the menu to the right of the monitor and selecting **Pause**. When you are ready to resume monitoring, click **Resume**.

You can copy a monitor by selecting **Duplicate** from the menu. You can then modify the new monitor and change the object/metric, filter, conditions, email recipients, etc.

If a monitor is no longer needed, you can delete it by selecting **Delete** from the menu.

Monitor Groups and System-Defined Monitors

Grouping allows you to view and manage related monitors. For example, you can have a monitor group dedicated to the storage in your environment, or monitors relevant to a certain recipient list.



Monitors are grouped in the following categories:

- **All Monitors**: All monitors, both user-created and system-defined
- **Custom Monitors**: Includes user-created monitors
- **System Monitor groups**: These groups include system-defined monitors, which are based on the data collectors in your environment
- **User-Created Monitor groups**: These are monitor groups you create

The number of monitors contained in a group is shown next to the group name.

> Custom monitors can be paused, resumed, deleted, or moved to another group. System-defined monitors can be paused and resumed but can not be deleted or moved.

System-Defined Monitors

System-defined monitors are comprised of pre-defined metrics and conditions, as well as default descriptions and corrective actions, which can not be modified. You *can* modify the notification recipient list for system-defined monitors. To view the metrics, conditions, description and corrective actions, or to modify the recipient list, open a system-defined monitor group and click the monitor name in the list.

System-defined monitor groups cannot be modified or removed.

The following system-defined monitors are available, in the noted groups.

- **ONTAP Infrastructure** includes monitors for infrastructure-related issues in ONTAP clusters.
- **ONTAP Workload Examples** includes monitors for workload-related issues.
- Monitors in both group default to *Paused* state.

Group	Monitor Name	Monitor Description
ONTAP Infrastructure	Storage Capacity Limit	When a storage pool (aggregate) fills up, I/O operations slow down and finally cease causing a storage outage incident. A warning alert indicates that planned action should be taken soon to restore minimum free space. A critical alert indicates that service disruption is imminent and emergency measures should be taken to free up space to ensure service continuity.
ONTAP Infrastructure	Storage Performance Limit	When a storage pool reaches its performance limit, operations slow down, latency goes up and workloads and applications may start failing. ONTAP evaluates the storage pool utilization due to workloads and estimates what percent of performance has been consumed. A warning alert indicates that planned action should be taken to reduce storage pool load as there may not be enough storage pool performance left to service workload peaks. A critical alert indicates that a performance brownout is imminent and emergency measures should be taken to reduce storage pool load to ensure service continuity.
ONTAP Infrastructure	Fiber Channel Port High Utilization	Fiber Channel Protocol ports are used to receive and transfer the SAN traffic between the customer host system and the ONTAP LUNs. If the port utilization is high, then it will become a bottleneck and it will ultimately affect the performance of Fiber Channel Protocol workloads. A warning alert indicates that planned action should be taken to balance network traffic. A critical alert indicates that service disruption is imminent and emergency measures should be taken to balance network traffic to ensure service continuity.

Group	Monitor Name	Monitor Description
ONTAP Infrastructure	Network Port High Utilization	Network ports are used to receive and transfer the NFS, CIFS, and iSCSI protocol traffic between the customer host systems and the ONTAP volumes. If the port utilization is high then it becomes a bottleneck and it ultimately affects the performance of NFS, CIFS, and iSCSI workloads. A warning alert indicates that planned action should be taken to balance network traffic. A critical alert indicates that service disruption is imminent and emergency measures should be taken to balance network traffic to ensure service continuity.
ONTAP Workload Examples	Volume Inodes Limit	Volumes that store files use index nodes (inode) to store file metadata. When a volume exhausts its inode allocation, no more files can be added to it. A warning alert indicates that planned action should be taken to increase the number of available inodes. A critical alert indicates that file limit exhaustion is imminent and emergency measures should be taken to free up inodes to ensure service continuity.
ONTAP Workload Examples	QTree Capacity is Full	A qtree is a logically defined file system that can exist as a special subdirectory of the root directory within a volume. Each qtree has a default space quota or a quota defined by a quota policy to limit amount of data stored in the tree within the volume capacity. A warning alert indicates that planned action should be taken to increase the space. A critical alert indicates that service disruption is imminent and emergency measures should be taken to free up space to ensure service continuity.

Group	Monitor Name	Monitor Description
ONTAP Workload Examples	Lun High Latency	LUNs are objects that serve the IO traffic often driven by performance sensitive applications such as databases. High LUN latencies means that the applications themselves may suffer and be unable to accomplish their tasks. A warning alert indicates that planned action should be taken to move the LUN to appropriate Node or Aggregate. A critical alert indicates that service disruption is imminent and emergency measures should be taken to ensure service continuity. The following are expected latencies based on media type - SSD up to 1-2 milliseconds; SAS up to 8-10 milliseconds and SATA HDD 17-20 milliseconds
ONTAP Workload Examples	NVMe Namespace High Latency	NVMe Namespaces are objects that serve the IO traffic often driven by performance sensitive applications such as databases. High NVMe Namespaces latencies means that the applications themselves may suffer and be unable to accomplish their tasks. A warning alert indicates that planned action should be taken to move the LUN to appropriate Node or Aggregate. A critical alert indicates that service disruption is imminent and emergency measures should be taken to ensure service continuity.
ONTAP Workload Examples	Qtree Files Soft Limit	A qtree is a logically defined file system that can exist as a special subdirectory of the root directory within a volume. Each qtree has a quota of the number of files that it can contain in order to maintain a manageable file system size within the volume. A qtree maintains a soft file number quota in order to be able to alert the user proactively before reaching the limit of files in the qtree and being unable to store any additional files. Monitoring the number of files within a qtree ensures that the user receives uninterrupted data service.

Group	Monitor Name	Monitor Description
ONTAP Workload Examples	Qtree Files Hard Limit	A qtree is a logically defined file system that can exist as a special subdirectory of the root directory within a volume. Each qtree has a quota of the number of files that it can contain in order to maintain a manageable file system size within the volume. A qtree maintains a hard file number quota beyond which new files in the tree are denied. Monitoring the number of files within a qtree ensures that the user receives uninterrupted data service.
ONTAP Workload Examples	QTree Capacity Soft Limit	A qtree is a logically defined file system that can exist as a special subdirectory of the root directory within a volume. Each qtree has a space quota measured in KBytes that it can use to store data in order to control the growth of user data in volume and not exceed its total capacity. A qtree maintains a soft storage capacity quota in order to be able to alert the user proactively before reaching the total capacity quota limit in the qtree and being unable to store data anymore. Monitoring the amount of data stored within a qtree ensures that the user receives uninterrupted data service.
ONTAP Workload Examples	QTree Capacity Hard Limit	A qtree is a logically defined file system that can exist as a special subdirectory of the root directory within a volume. Each qtree has a space quota measured in KBytes that it can use to store data in order to control the growth of user data in volume and not exceed its total capacity. A qtree hard storage capacity quota serves as a threshold beyond which the writes on the qtree are denied. Monitoring the amount of data stored within a qtree ensures that the user receives uninterrupted data service.

Group	Monitor Name	Monitor Description
ONTAP Workload Examples	User Quota Capacity Soft Limit	ONTAP recognize the users of Unix or Windows systems that have the rights to access volumes, files or directories within a volume. As a result, ONTAP allows the customers to configure storage capacity for their users or groups of users of their Linux or Windows systems. The user or group policy quota limits the amount of space the user can utilize for their own data. A soft limit of this quota allows proactive notification of the user when the amount of capacity used within the volume is reaching the total capacity quota. Monitoring the amount of data stored within a user or group quota ensures that the user receives uninterrupted data service.
ONTAP Workload Examples	User Quota Capacity Hard Limit	ONTAP recognize the users of Unix or Windows systems that have the rights to access volumes, files or directories within a volume. As a result, ONTAP allows the customers to configure storage capacity for their users or groups of users of their Linux or Windows systems. The user or group policy quota limits the amount of space the user can utilize for their own data. A hard limit of this quota allows notification of the user when the amount of capacity used within the volume is right before reaching the total capacity quota. Monitoring the amount of data stored within a user or group quota ensures that the user receives uninterrupted data service.

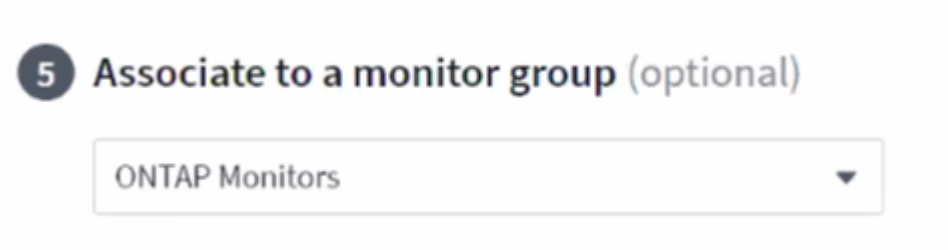
Group	Monitor Name	Monitor Description
ONTAP Workload Examples	Volume High Latency	Volumes are objects that serve the IO traffic often driven by performance sensitive applications including devOps applications, home directories, and databases. High volume latencies means that the applications themselves may suffer and be unable to accomplish their tasks. Monitoring volume latencies is critical to maintain application consistent performance. The following are expected latencies based on media type - SSD up to 1-2 milliseconds; SAS up to 8-10 milliseconds and SATA HDD 17-20 milliseconds
ONTAP Workload Examples	Snapshot Reserve Capacity is Full	Storage capacity of a volume is necessary to store application and customer data. A portion of that space, called snapshot reserved space, is used to store snapshots which allow data to be protected locally. The more new and updated data stored in the ONTAP volume the more snapshot capacity is used and less snapshot storage capacity will be available for future new or updated data. If the snapshot data capacity within a volume reaches the total snapshot reserve space it may lead to the customer being unable to store new snapshot data and reduction in the level of protection for the data in the volume. Monitoring the volume used snapshot capacity ensures data services continuity.
ONTAP Workload Examples	Volume Capacity is Full	Storage capacity of a volume is necessary to store application and customer data. The more data stored in the ONTAP volume the less storage availability for future data. If the data storage capacity within a volume reaches the total storage capacity may lead to the customer being unable to store data due to lack of storage capacity. Monitoring the volume used storage capacity ensures data services continuity.

Custom Monitor Groups

To create a new custom monitor group, click the **"+" Create New Monitor Group** button. Enter a name for the group and click **Create Group**. An empty group is created with that name.

To add monitors to the group, go to the *All Monitors* group (recommended) and do one of the following:

- To add a single monitor, click the menu to the right of the monitor and select *Add to Group*. Choose the group to which to add the monitor.
- Click on the monitor name to open the monitor's edit view, and select a group in the *Associate to a monitor group* section.



Remove monitors by clicking on a group and selecting *Remove from Group* from the menu. You can not remove monitors from the *All Monitors* or *Custom Monitors* group. To delete a monitor from these groups, you must delete the monitor itself.

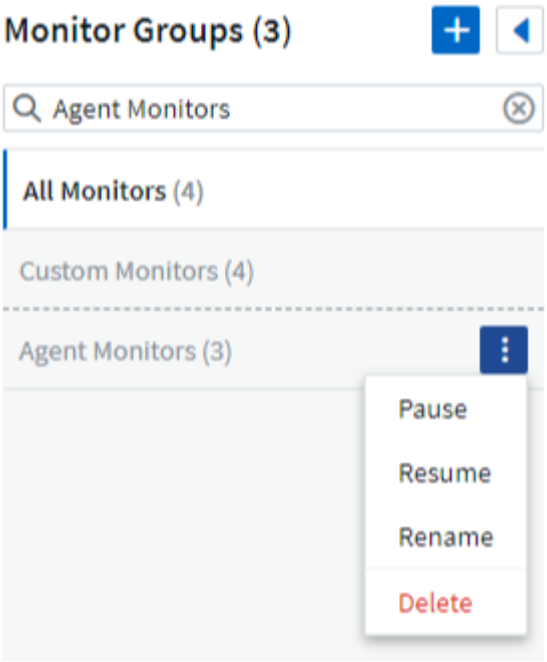
> Removing a monitor from a group does not delete the monitor from Cloud Insights. To completely remove a monitor, select the monitor and click *Delete*. This also removes it from the group to which it belonged and it is no longer available to any user.

You can also move a monitor to a different group in the same manner, selecting *Move to Group*.

> Each monitor can belong to only a single group at any given time.

To pause or resume all monitors in a group at once, select the menu for the group and click *Pause* or *Resume*.

Use the same menu to rename or delete a group. Deleting a group does not delete the monitors from Cloud Insights; they are still available in *All Monitors*.



More Information

- Viewing and Dismissing Alerts

Viewing and Managing Alerts from Monitors

Cloud Insights displays alerts when monitored thresholds are exceeded.

Monitors and Alerting is available in Cloud Insights Standard Edition and higher.

Viewing and Managing Alerts

To view and manage alerts, do the following.

1. Navigate to the **Alerts > Integration Alerts** page.
2. A list of up to the most recent 1,000 alerts is displayed. You can sort this list on any field by clicking the column header for the field. The list displays the following information. Note that not all of these columns are displayed by default. You can select columns to display by clicking on the "gear" icon :
   - **Alert ID**: System-generated unique alert ID
   - **Triggered Time**: The time at which the relevant Monitor triggered the alert
   - **Current Severity** (Active alerts tab): The current severity of the active alert
   - **Top Severity** (Resolved alerts tab); The maximum severity of the alert before it was resolved
   - **Monitor**: The monitor configured to trigger the alert
   - **Triggered On**: The object on which the monitored threshold was breached
   - **Status**: Current alert status, *New* or *In Process*
   - **Active Status**: *Active* or *Resolved*
   - **Condition**: The threshold condition that triggered the alert

- **Metric**: The object's metric on which the monitored threshold was breached
- **Monitor Status**: Current status of the monitor that triggered the alert

You can manage an alert by clicking the menu to the right of the alert and choosing one of the following:

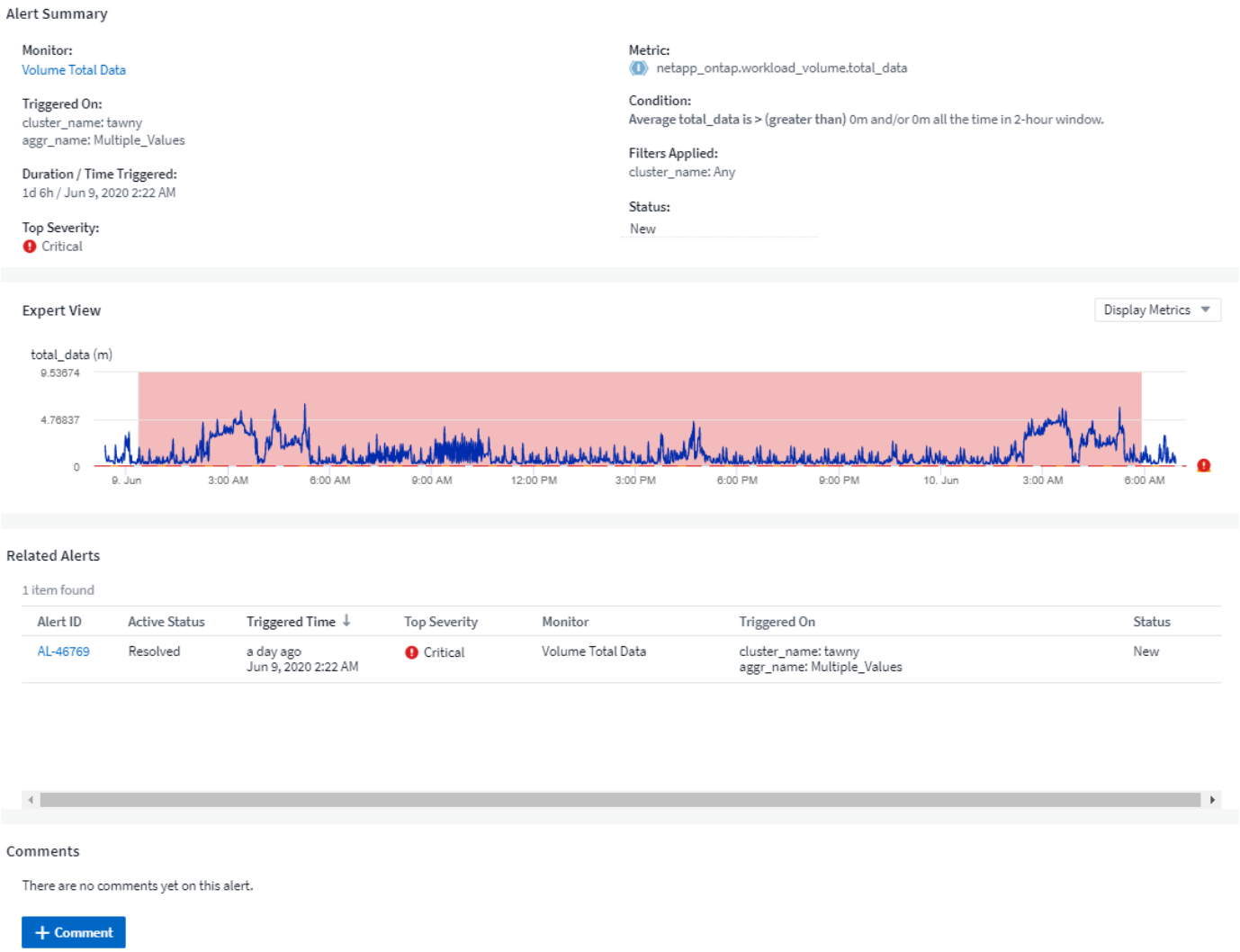
- **In Process** to indicate that the alert is under investigation or otherwise needs to be kept open
- **Dismiss** to remove the alert from the list of active alerts.

You can manage multiple alerts by selecting the checkbox to the left of each Alert and clicking *Change Selected Alerts Status*.

Clicking on an Alert ID opens the Alert Detail Page.

Alert Detail Page

The Alert Detail Page provides additional detail about the alert, including a *Summary*, an *Expert View* showing graphs related to the object's data, any *Related Assets*, and *Comments* entered by alert investigators.



"Permanently Active" Alerts

It is possible to configure a monitor in such a way for the condition to **always** exist on the monitored object—for example, IOPS > 1 or latency > 0. These are often created as 'test' monitors and then forgotten. Such monitors

create alerts that stay permanently open on the constituent objects, which can cause system stress and stability issues over time.

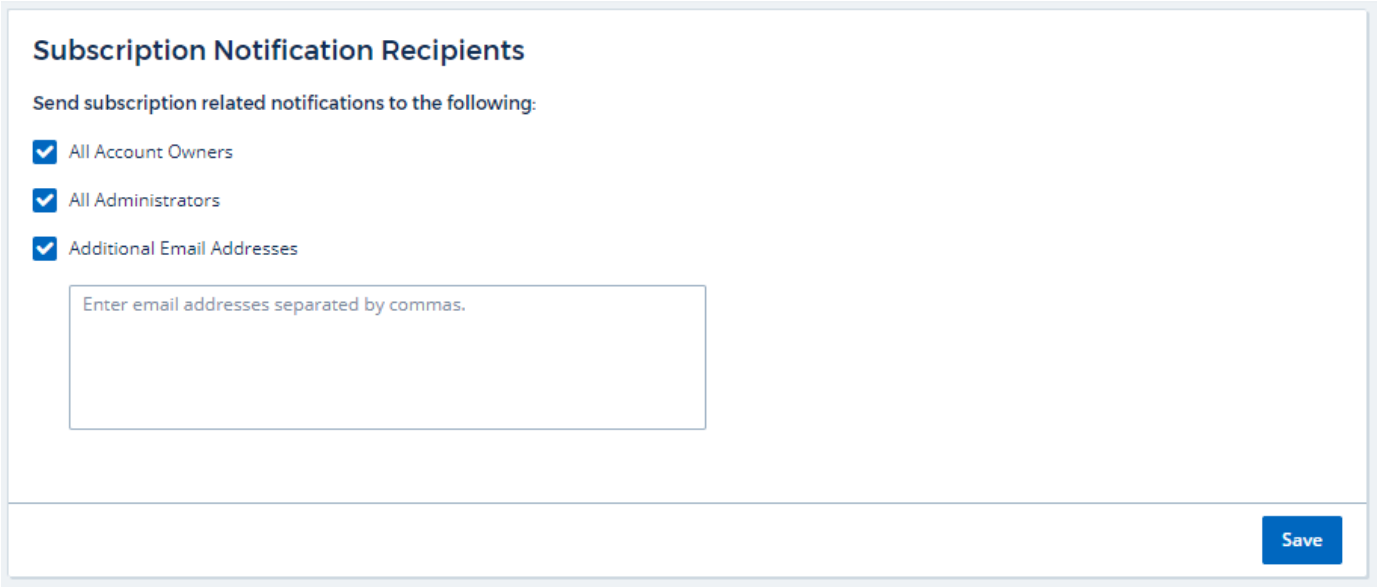
To prevent this, Cloud Insights will automatically close any "permanently active" alert after 7 days. Note that the underlying monitor conditions may (probably will) continue to exist, causing a new alert to be issued almost immediately, but this closing of "always active" alerts alleviates some of the system stress that can otherwise occur.

Configuring Email Notifications

You can configure an email list for subscription-related notifications, as well as a global email list of recipients for notification of performance policy threshold violations.

To configure notification email recipient settings, go to the **Admin > Notifications** page.

Subscription Notification Recipients



To configure recipients for subscription-related event notifications, go to the "Subscription Notification Recipients" section.
You can choose to have email notifications sent for subscription-related events to any or all of the following recipients:

- All Account Owners
- All Administrators
- Additional Email Addresses that you specify

The following are examples of the types of notifications that might be sent, and user actions you can take.

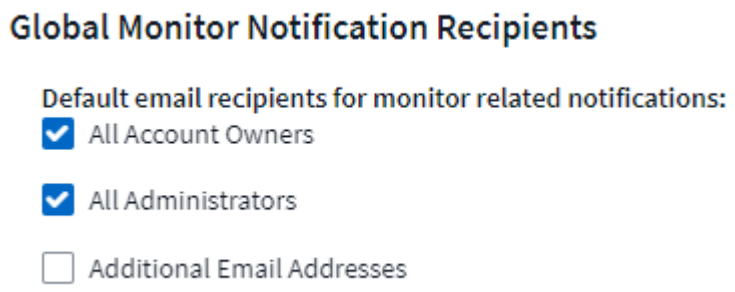
Notification:	User Action:
Trial or subscription has been updated	Review subscription details on the Subscription page
Subscription will expire in 90 days Subscription will expire in 30 days	No action needed if “Auto Renewal” is enabled Contact NetApp sales to renew the subscription

Trial ends in 2 days	Renew trial from the Subscription page. You can renew a trial one time. Contact NetApp sales to purchase a subscription
Trial or subscription has expired Account will stop collecting data in 48 hours Account will be deleted after 48 hours	Contact NetApp sales to purchase a subscription

Global Recipient List for Alerts

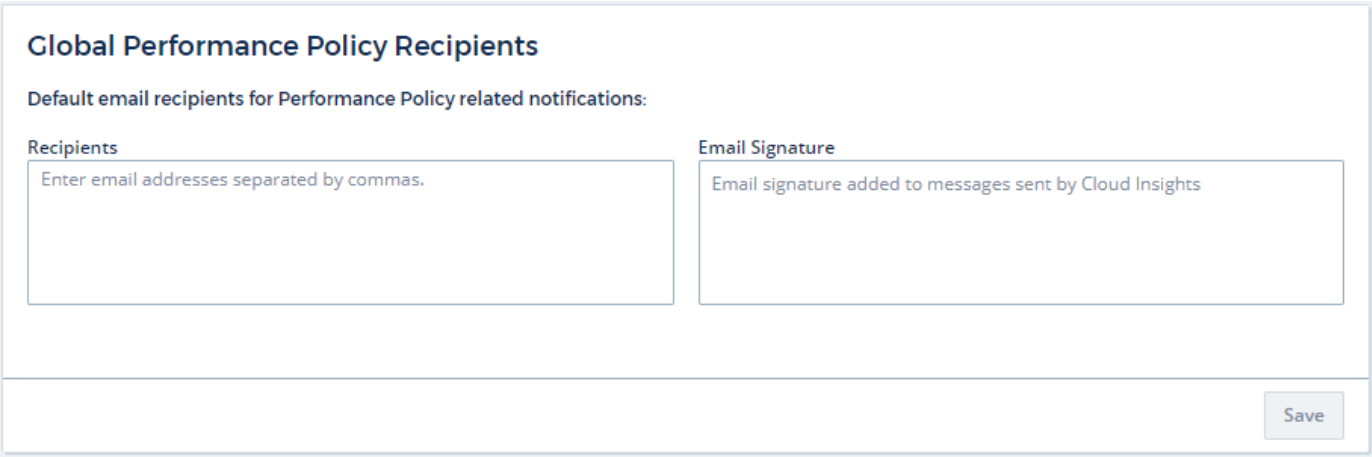
Email notifications of alerts are sent to the alert recipient list for every action on the alert. You can choose to send alert notifications to a global recipient list.

To configure global alert recipients, click on **Admin > Notifications** and choose the desired recipients in the **Global Monitor Notification Recipients** section.



You can always override the global recipients list for an individual monitor when creating or modifying the monitor.

Global Recipient List for Performance Policy Notifications



To add recipients to the global performance policy notification email list, go to the "Global Performance Policy Recipients" section and enter email addresses separated by commas. Emails sent as alerts from performance policy threshold violations will be sent to all recipients on the list.

If you make a mistake, you can click on [x] to remove a recipient from the list.

You can also add an optional signature block that will be attached to the email notifications sent.

You can override the global list for a specific policy when you configure that policy.

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